

**A
ON JOB TRAINING (OJT) REPORT
AS
“ AI ML Intern” AT
CodeIt InfoTech PVT. LTD.**

SUBMITTED TO THE



**PUNE DISTRICT EDUCATION ASSOCIATION'S
PROF. RAMKRISHNA MORE ARTS, COMMERCE AND SCIENCE
COLLEGE (AUTONOMOUS) AKURDI, PUNE-411044**

**FOR THE DEGREE OF
T. Y. BCA (SCIENCE) SEM-VI**

**BY
ROHAN AMITIGIRI GOSWAMI
Roll No. 131
Seat No. S129724**

**UNDER THE GUIDANCE OF
PROF. KALYANI GHOLAP**

**DEPARTMENT OF BCA SCIENCE
PUNE DISTRICT EDUCATION ASSOCIATION'S
PROF. RAMKRISHNA MORE ARTS, COMMERCE & SCIENCE
COLLEGE (AUTONOMOUS), AKURDI, PUNE-411 044**

YEAR 2025-26

CERTIFICATE

This is certified that Mr Rohan Amitgiri Goswami, bearing the Roll No 131 and Exam Seat No S129724 of T.Y.BCA(Science) Sem-VI, has successfully completed BSOJP-361 On Job Training (OJT) entitled AI ML Intern at CodeIT InfoTech Pvt. Ltd. during the academic year 2025-2026.

Prof. Kalyani Gholap
Project Guide

Prof. Shital Deshmukh
Head of Department

Internal Examiner

External Examiner



CODEIT Pvt. Ltd.

Office No.03,3rd Floor,Mata Rajeshwari Complex, Opp. Manjari Stud Farm, Near New Golden Bakery, Shewalewadi, Pune - 412 307

Internship Offer Letter

Date:13-December-2025

To,

Candidate Name: Rohan Goswami

Subject: Internship Offer - AZ/ML

Dear Rohan Goswami

We are pleased to offer you an Internship in
AZ/ML at CODEIT Pvt. Ltd.

Duration: 45 Days

Start Date: 22-Dec-2025

Mode: online (Online / Offline / Hybrid)

You will work on practical assignments under professional guidance. Upon successful completion, an Internship Certificate will be issued.

This internship does not guarantee employment and is subject to company policies.

Kindly confirm your acceptance.

Regards,



Preeti Kalra

Founder & Director

CODEIT INFOTECH

CODEIT Infotech | CIN: U62099PN2025OPC248149 | www.codeitinfotech.com |
careers@codeitinfotech.com
Office No. 03, 3rd Floor, Mata Rajeshwari Complex, Manjari Stud Farm, Near Golden
Bakery, Shewalwadi, Pune – 412 307

INTERNSHIP COMPLETION CERTIFICATE

This is to certify that. **Rohan Goswami** has
successfully completed a 45 Days Internship in Artificial
Intelligence & Machine Learning (AI/ML) at CODE IT
Infotech.

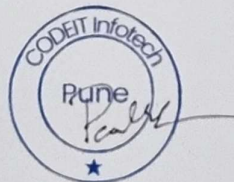
During the internship, the candidate demonstrated sound
technical knowledge in Python and Machine Learning
concepts and successfully completed the assigned
project work. Their performance and conduct were found
to be professional and satisfactory throughout the
program.

We wish them every success in their future academic and
professional endeavors.

Date: 28/02/2026

Place: Pune

For CODE IT Infotech



PREETI KALRA
FOUNDER, CODEIT INFOTECH



Pune District Education Association's
Prof. Ramkrishna More Arts, Commerce and Science College
(Autonomous)
Akurdi, Pune-411044
Department of Computer Science

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Acknowledgement

The internship opportunity I had with CodeIT InfoTech Pvt. Ltd. was a valuable learning experience for me. I am grateful for having the opportunity to work in a professional environment and to interact with knowledgeable individuals who guided me throughout this internship period.

I would like to express my sincere gratitude to **Mr. Lakhan Jadhav**, my mentor and supervisor at CodeIT InfoTech Pvt. Ltd., for his constant guidance, support, and encouragement. His insights and willingness to share knowledge helped me stay on the right path and successfully complete my internship tasks.

I extend my deepest gratitude to the Honourable Principal, **Dr. A. J. Khandagle Sir**, of Prof. Ramkrishna More Arts, Commerce and Science College (Autonomous), Pune, for providing me with the opportunity to undertake and complete this internship. I would also like to thank **Prof. Shital Deshmukh**, Head of the Department of Computer Science, and **Prof. Kalyani Gholap**, my project guide, for their valuable guidance, encouragement, and support throughout the internship.

I am equally thankful to all the faculty members, my friends, and colleagues who directly or indirectly contributed to the successful completion of this internship. I will continue to strive for improvement and apply the knowledge and experience gained during this internship to achieve my future career goals.

Rohan Amitgiri Goswami
TY BCA (Science)

Company Profile

CodeIT InfoTech (OPC) Pvt. Ltd. is an industry-focused IT training and software development organization based in Pune, Maharashtra, India. The company provides hands-on training in programming, advanced technologies, and emerging domains such as Artificial Intelligence and Data Science.

The organization is committed to delivering practical, skill-oriented education aligned with current industry requirements. With experienced trainers and strong industry exposure, CodeIT InfoTech focuses on enhancing students' technical capabilities through real-time projects, internships, and structured training programs. It aims to bridge the gap between academic learning and industry expectations by equipping learners with practical knowledge and problem-solving skills.

Company Overview

- **Company Name:** CodeIT InfoTech (OPC) Pvt. Ltd.
- **CIN:** U62099PN2025OPC248149
- **Industry:** IT Training & Software Development
- **Head Office:** Pune, Maharashtra, India
- **Office Address:** 3rd Floor, Mata Rajeshwari Complex, Solapur - Pune Hwy, opp. Manjri Stud Farm, Shewalewadi, Pune, Maharashtra – 412307
- **Website:** <https://codeitsoftwaretraining.in/>
- **Official Email:** codeit@codeitsoftwaretraining.com
- **Contact Number:** +91 7231925496

Services and Capabilities

- Training in Programming Languages (Python, Java, etc.)
- Artificial Intelligence & Machine Learning Training
- Data Science and Data Analytics
- Full Stack Web Development
- Real-time Project Development
- Internship Programs for Students
- Placement Assistance and Career Guidance
- Industry-Oriented Skill Development Programs

Company Objectives

- To provide industry-aligned and practical IT education
- To bridge the gap between academic learning and industry requirements
- To enhance technical and problem-solving skills of students
- To promote hands-on learning through real-time projects and internships
- To support students in building successful careers in the IT industry

ABSTRACT

This abstract presents the technical design, implementation, and execution of a real-time Face Recognition System developed during an AI/ML internship at CodeIT InfoTech (OPC) Pvt. Ltd.. The project focuses on building a lightweight computer vision pipeline capable of detecting, encoding, and recognizing human faces from live webcam input using pre-trained deep learning models. It demonstrates the practical application of machine learning concepts, image processing techniques, and real-time system optimization within a structured development workflow.

The core of the system is based on feature extraction using a pre-trained deep metric learning model, which converts facial images into 128-dimensional numerical embeddings. These embeddings enable efficient similarity comparison using distance-based metrics, allowing the system to classify faces as known or unknown. A threshold-based decision mechanism was implemented to balance recognition accuracy and false detection rates, ensuring reliable performance under controlled conditions.

A significant aspect of the implementation involved designing a lightweight and scalable data pipeline. The system integrates an image dataset with structured JSON metadata to map identities such as name, age, and unique ID. The architecture ensures efficient data loading, encoding, and retrieval, forming a bridge between static data storage and dynamic real-time inference. Emphasis was placed on maintaining data integrity by validating input images and handling edge cases such as missing or multiple faces.

The project also incorporates real-time processing optimization techniques to ensure smooth performance on CPU-based systems. By applying frame downscaling and minimizing redundant computations, the system achieves efficient face detection and recognition within live video streams. Additionally, robust error handling mechanisms were implemented to manage camera failures, invalid inputs, and safe termination of the application, enhancing system reliability.

In conclusion, this work represents the integration of machine learning, computer vision, and software engineering principles to develop a practical, real-time recognition system. The project highlights the ability to translate theoretical AI concepts into a functional application, demonstrating problem-solving skills, system design capabilities, and readiness to contribute to real-world intelligent systems and emerging technological solutions.

INTRODUCTION

With the rapid advancement of Artificial Intelligence and Computer Vision, face recognition technology has become an important component in modern digital systems. It is widely used in applications such as security systems, attendance management, access control, and user authentication. Face recognition systems provide a contactless and efficient method of identifying individuals, making them highly relevant in today's technology-driven environment.

The objective of this project is to design and develop a real-time Face Recognition System capable of detecting and identifying individuals from a live video stream. The system is implemented using Python and leverages libraries such as OpenCV and `face_recognition`, which utilize pre-trained deep learning models for facial feature extraction. The approach focuses on converting facial images into numerical representations (embeddings) and comparing them to identify known individuals.

The system follows a structured pipeline that includes image acquisition, face detection, feature extraction, and identity matching. A dataset of labeled images is used to generate facial encodings, which are then compared with real-time inputs captured through a webcam. Based on similarity measures and a predefined threshold, the system classifies faces as known or unknown.

In addition to recognition, the system integrates a lightweight data management approach using JSON files to store user-related information such as name, age, and identification details. This ensures efficient mapping between facial data and user metadata. The implementation also focuses on optimizing performance for real-time execution on standard hardware by applying techniques such as frame resizing and efficient computation.

Furthermore, the system is designed with robustness in mind, incorporating validation checks and error handling mechanisms to manage scenarios such as missing faces in images, invalid inputs, and camera failures. This enhances the reliability and usability of the application in practical environments.

Overall, this project demonstrates the application of machine learning and computer vision concepts in building a real-time system, providing hands-on experience in developing intelligent applications and understanding real-world implementation challenges.

OBJECTIVES

The primary objective of this internship was to develop a comprehensive understanding of Artificial Intelligence and Computer Vision by bridging theoretical concepts with real-world implementation. The program was designed to enable practical exposure to building intelligent systems capable of performing real-time face detection and recognition using machine learning techniques.

1. Implementation of Face Recognition Pipeline

The core objective was to design and implement an end-to-end face recognition system. This involved developing a structured pipeline consisting of face detection, feature extraction, encoding, and identity matching. By working with libraries such as OpenCV and face_recognition, the goal was to understand how computer vision systems process and interpret visual data in real-time environments.

2. Feature Extraction and Embedding Generation

A key objective was to utilize pre-trained deep learning models to extract meaningful facial features and convert them into 128-dimensional embeddings. This phase focused on understanding how deep metric learning enables similarity-based recognition rather than traditional classification approaches, ensuring efficient and scalable identity matching.

3. Similarity Measurement and Decision Logic

Another important objective was to implement a reliable recognition mechanism using distance-based similarity metrics such as Euclidean distance. By introducing a threshold-based classification system (0.50), the goal was to balance accuracy and reliability while minimizing false positives and incorrect identifications.

4. Design of Data Handling and Integration

The project aimed to develop a lightweight and efficient data management system by integrating an image dataset with structured JSON metadata. This objective focused on ensuring proper identity mapping (Name, Age, ID), maintaining data consistency, and enabling smooth interaction between stored data and real-time recognition processes.

5. Optimization for Real-Time Performance

A significant objective was to ensure smooth execution of the system in real-time environments. This involved applying optimization techniques such as frame downscaling (0.25×) and minimizing computational overhead to achieve efficient processing on CPU-based systems without compromising functionality.

6. Robustness and Error Handling

The system was designed with an emphasis on reliability by handling edge cases such as missing faces in images, multiple face detections, and camera initialization failures. This objective ensured that the application remains stable and performs consistently under different runtime conditions.

7. Practical Exposure to AI System Development

The overall objective was to gain hands-on experience in designing, developing, and optimizing a real-time AI-based system. This included understanding the challenges of performance, data quality, and system integration, thereby preparing for real-world applications in computer vision and intelligent systems.

Internship Undertaking

1. Student Name	Rohan Amitgiri Goswami		
2. Current Address	G3 Kriplani Bhavan Pimpri Colony, near Pimpri Colony Post-Office, Pimpri Pune - 411017		
3. Residence Address	G3 Kriplani Bhavan Pimpri Colony, near Pimpri Colony Post-Office, Pimpri Pune – 411017		
4. Email id	rohangoswami1134@gmail.com		
5. Mobile No.	+91 7769918315		
6. Aadhar	4789 6629 9662		
7. PAN	DXHPG5787B		
8. Overall GPA	9.68		
9. Mode of Internship	Online		
10. Internship Preferences			
	Location	Core Area	Organization / Institute
Preference-1	Pune	Artificial Intelligence and Machine Learning	CodeIT InfoTech Pvt. Ltd.
Preference-2	Online Pune	Data Science	Shamgar Software Solutions
Preference-3	Pune	Data Science and Machine Learning	Any IT Company
confirm that I agree with the terms, conditions, and requirements of the Internship Policy Student Signature: Date: I confirm that the student has attended the internship orientation and has met all paperwork and process requirements to participate in the internship program, and has received approval from his/her mentor. Sign of Department Faculty Coordinator: Date :			

RESUME

ROHAN GOSWAMI

+91-7769918315 | rohangoswami1134@gmail.com

Pune, Maharashtra

[LinkedIn](#)

[GitHub](#)

Career Objective

Data Science enthusiast with hands-on experience in Machine Learning, Deep Learning, and Computer Vision. Skilled in building end-to-end data pipelines, performing exploratory data analysis (EDA), feature engineering, and deploying scalable models. Seeking to leverage data-driven solutions to solve real-world business problems.

Technical Skills

- Programming – Python, SQL (PostgreSQL, MySQL, Microsoft SQL Server)
- Libraries – Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, NLTK
- Machine Learning – Feature Engineering, Hyperparameter Tuning, Cross Validation, Model evaluation
- Deep Learning – ANN, CNN, RNN, Specialized Architectures, Transfer Learning, Model optimization
- Natural Language Processing – Text processing, Tokenization, TF-IDF, Transformers, Sentiment analysis
- Computer Vision – Image Processing, Image Classification, Object Detection, Feature Extraction
- Model Deployment – FlaskAPI, FastAPI, Streamlit, Amazon Web Services(AWS), Google Cloud Engine

Soft Skills

- Analytical thinking
- Communication
- Problem Solving

Projects

LAPTOP PRICE PREDICTION – [Project Link](#)

- Developed an end-to-end machine learning pipeline to predict laptop prices using 1,300+ real-world records.
- Performed data cleaning and exploratory data analysis (EDA) to identify key pricing factors.
- Engineered 13+ features including HDD/SSD decomposition, CPU/GPU categorization, and pixel density (PPI).
- Implemented preprocessing using ColumnTransformer, OneHotEncoder, and feature scaling.
- Trained and optimized multiple regression models using cross-validation and hyperparameter tuning.
- Improved prediction performance using ensemble techniques (Voting & Stacking Regressors).
- Achieved R^2 score of 88.7% and MAE of 0.158 (log scale).
- Deployed the model using Streamlit for real-time predictions.

Work Experience

AI/ML INTERN – CODEIT PVT.LTD – PUNE

December 2025

- Developed a real-time face recognition system using Python, OpenCV, and dlib for live identity detection.
- Generated 128-dimensional facial embeddings and implemented threshold-based classification for accurate recognition.
- Built an efficient pipeline integrating image datasets with structured JSON metadata for identity mapping.
- Optimized system performance by ~40% using frame downscaling, enabling smooth execution on CPU-only systems.
- Implemented robust error handling for missing faces, dataset inconsistencies, and camera failures.
- Applied computer vision techniques to improve detection reliability under real-time conditions.

Education

BACHELOR OF COMPUTER APPLICATION – Prof. Ramkrishna More Arts, Commerce & Science College – Pune

CGPA: 9.40

March-2023 to 2026

Student Diary (Log) Recording

Week	Date	Task Assigned	Activities Performed	Key Learnings	Additional Remarks
1	27/12/2025 – 02/01/2026	Introduction to AI/ML	Learned basics of AI, ML and Python Libraries	Understanding of AI/ ML Concepts	Strong Foundations built
2	03/01/2026 – 09/01/2026	Data Handling	Worked on how to gather data from different formats and required data preprocessing	Data Integration and Cleaning	Awareness about data preprocessing
3	10/01/2026 – 16/01/2026	ML Algorithms	Applied various Regression and Classification models	Internal Working and Implementation of models.	Clear understanding of models working
4	17/01/2026 – 23/01/2026	Model Evaluation And Hyperparameter Tuning	Evaluated various regression and classification model with appropriate metrics and improved them using Hyperparameter tuning	Learned to evaluate model based on requirements and how hyperparameter tuning can give a actual idea of model performance	Hyperparameter tuning is always a must step in developing a good ML model
5	24/01/2026 – 30/01/2026	NLP Concepts - Introduction and Application	Performed text preprocessing steps and applied NLP concepts like Stemming, Lemmetization and Tokenization	NLP Important Concepts and Real world application	New subfield of ML
6	31/012026 – 06/02/2026	Computer Vision and Final Project Initialization	Learned computer vision basics and started face recognition project, including dataset loading and face encoding.	Understanding of face detection and feature extraction using pre-trained models.	Project development initiated successfully
7	07/02/2026 – 13/02/2026	Project Complition	Completed real-time face recognition system with webcam, matching logic, and visualization.	Real-time implementation and similarity-based recognition concepts.	Project completed successfully

Attendance Sheet

Name & Address of Organization

CodeIt InfoTech Pvt.Ltd

3rd Floor, Mata Rajeshwari Complex, Solapur - Pune Hwy, opp. Manjri Stud Farm,
Shewalewadi, Pune, Maharashtra – 412307

Name of the Student	Rohan Amitgiri Goswami
Roll Number	131
Name of Role	AI/ML Intern
Date of Commencement of Training	27 th December 2025
Date of Completion of Training	10 th February2026

Week	Mon	Thu	Wed	Thu	Fri	Sat
1	P	P	P	P	P	P
2	P	P	P	P	P	A
3	P	P	P	P	P	P
4	P	P	P	P	P	P
5	P	p	p	p	P	p
6	p	p	A	p	p	p
7	p	p	p	-	-	-

- Attendance Sheet should remain affixed in Daily Training Diary. Do not remove or tear it off.
- Holidays should be marked in Red Ink in attendance column. Absent should be marked as 'A' in Red Ink.

Name and Signature with date of Internship Supervisor:

Supervisor or Evaluation of Intern

Student Name: Rohan Amitgiri Goswami

Date: 26 February 2026

Work Supervisor: Mrs. Preeti Kalra

Title: Founder & Director

Organization: CodeIt InfoTech Pvt. Ltd

Internship Address: Shewalewadi, Pune, Maharashtra – 412307

Dates of Internship: 27 December 2025 To 10 February 2026

Parameters	Needs Improvement	Satisfactory	Good	Excellent
Behaviors				✓
Performs in a dependable manner				✓
Cooperates with co-workers and supervisors				✓
Shows interest in work				✓
Learns quickly				✓
Shows initiative				✓
Produces high quality work				✓
Accepts responsibility			✓	
Accepts criticism			✓	
Demonstrates organizational skills			✓	
Uses technical knowledge and expertise				✓
Shows good judgment			✓	
Demonstrates creativity/originality				✓
Analyzes problems effectively				✓
Is self-reliant				✓
Communicates well				
Writes effectively			✓	
Has a professional attitude			✓	
Gives a professional appearance			✓	
Is punctual				✓
Uses time effectively				✓

Overall performance of student intern (circle one): (Needs improvement / Satisfactory / Good / Excellent)



Signature of Industry supervisor

Feedback of Internship Student

(To be filled by Students after Internship completion)

Student Name: Rohan Goswami Date: _____

Industrial Supervisor: Mrs. Preeti Kalra

Title: Director Founder

Supervisor Email: codeit@codeitsoftwaretraining.com

Internship is: Unpaid

Organization: CodeIt InfoTech Pvt. Ltd.

Internship Address: Pune

Faculty Coordinator: Prof. Kalyani Gholap

Department: BCA (Science)

Dates of Internship: 27th December 2025 To 23rd February 2026

Give a brief description of your internship work (title and tasks for which you were responsible):

- Worked as an **AI/ML Intern** at CodeIT InfoTech Pvt. Ltd., where I developed a real-time face recognition system using Python, OpenCV, and face_recognition.
- Handled dataset preparation, face encoding, similarity-based matching, and real-time webcam integration with labeled output visualization.

Was your internship experience related to your major area of study?

- Yes, to a large degree ✓
- Yes, to a slight degree
- No, not related at all

- Indicate the degree to which you agree or disagree with the following statements.

Experience has:	Strongly Agree	Agree	No opinion	Disagree	Strongly Disagree
Given me the opportunity to explore a career field		✓			
Allowed me to apply classroom theory to Practice	✓				
Helped me develop my decision-making and problem-solving skills	✓				
Expanded my knowledge about the work world prior to permanent employment	✓				
Helped me develop my written and oral communication skills		✓			
Provided a chance to use leadership skills (influence others, develop ideas with others, stimulate decision-making and action)		✓			
Expanded my sensitivity to the ethical implications of the work involved	✓				
Made it possible for me to be more confident in new situations	✓				
Given me a chance to improve my interpersonal skills	✓				
Helped me learn to handle responsibility and use my time wisely	✓				
Helped me discover new aspects of myself that I didn't know existed before		✓			
Helped me develop new interests and Abilities			✓		
Helped me clarify my career goals		✓			
Provided me with contacts which may lead to future employment	✓				
Allowed me to acquire information and/ or use equipment not available at my Institute			✓		

- In the Institute internship program, faculty members are expected to be mentors for students. Do you feel that your faculty coordinator served such a function?

Why or why not?

Ans: Yes, my faculty coordinator acted as a mentor by providing guidance, support, and timely feedback throughout the internship. Their suggestions helped me stay on track and improve my understanding of concepts.

- How well were you able to accomplish the initial goals, tasks and new skills that were set down in your learning contract? In what ways were you able to take a new direction or expand beyond your contract? Why were some goals not accomplished adequately?

Ans: I was able to accomplish most of the initial goals and tasks successfully. I also explored additional areas such as real-time system optimization beyond the planned objectives. Some advanced improvements were limited due to time constraints.

- What areas did you most develop and improve?

Ans: I improved my skills in computer vision, problem-solving, and real-time system implementation. I also developed a better understanding of handling data and building practical AI-based applications.

- What has been the most significant accomplishment or satisfying moment of your internship?

Ans: The most satisfying moment was successfully building and running the real-time face recognition system with accurate detection and labeling.

- What did you dislike about the internship?

Ans: There was nothing significant to dislike; however, more time for advanced features and improvements would have been beneficial.

- Considering your overall experience, how would you rate this internship? (Circle one). Satisfactory/ Good/ Excellent

Ans: Excellent

- Give suggestions as to how your internship experience could have been improved.

Ans: The internship could be improved by providing more time for project development and including additional hands-on sessions or discussions for deeper understanding. More real-world case studies would also be helpful.

Name: Rohan Amitgiri Goswami

Roll number: 131

Date:

Signature of Student

Evaluation of Internship by Institute

1. Name of Student _____
2. Mob. No. _____
3. Roll No. _____
4. Branch/Semester _____
5. Period of Training _____
6. Home Address with contact No. _____
7. Address of Training Site: _____
8. Address of Training Providing Agency: _____
9. Name/Designation of Training In- charge _____
10. Type of Work _____
11. Date of Evaluation _____

12. Please rate the following:

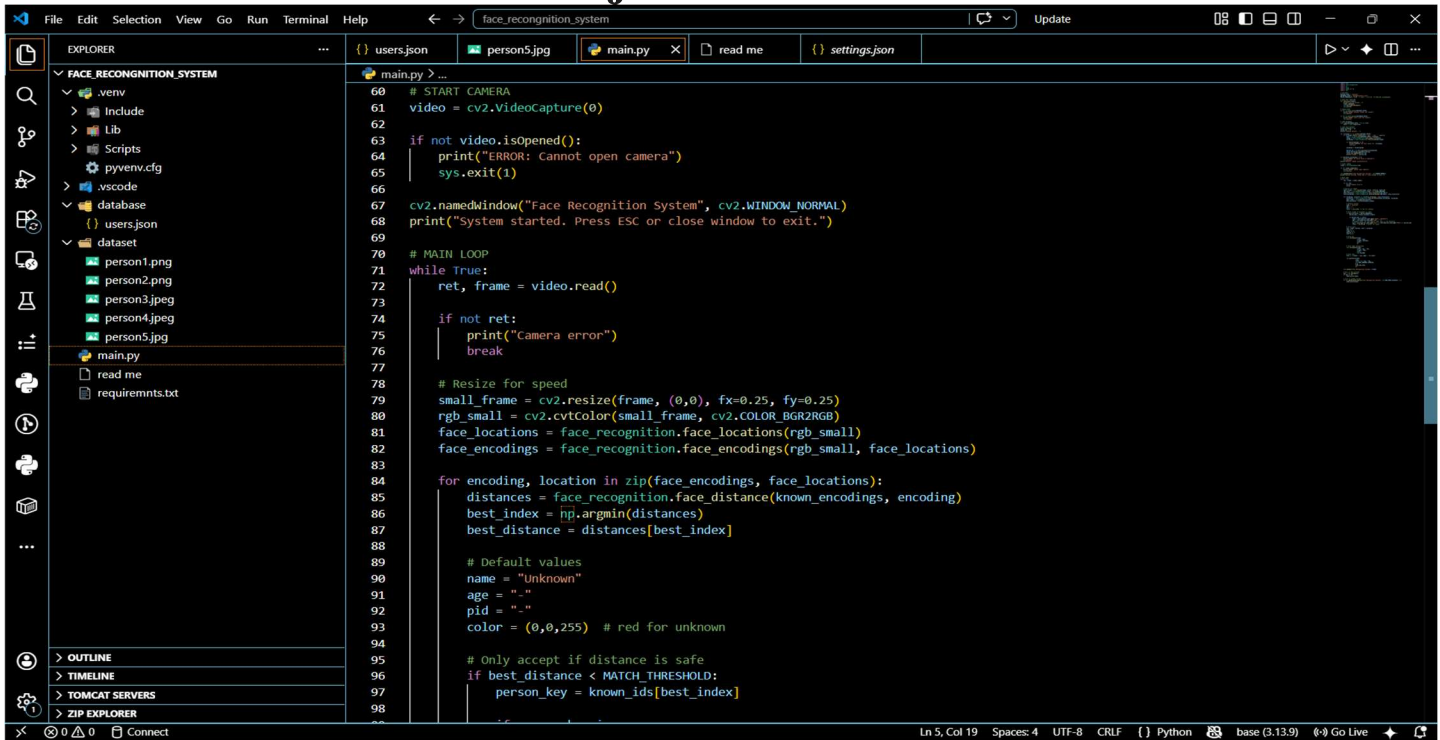
S.no	Particular	Grade
1	Quality and effectiveness of presentation	
2	Depth of knowledge and demonstrated skills	
3	Variety and relevance of learning experience	
4	Practical applications and relationships with concepts taught	
5	Internship Report	
6	Attendance record, student log, supervisor evaluation	

Overall grade: _____

Additional Remarks:

Signature of Faculty mentor

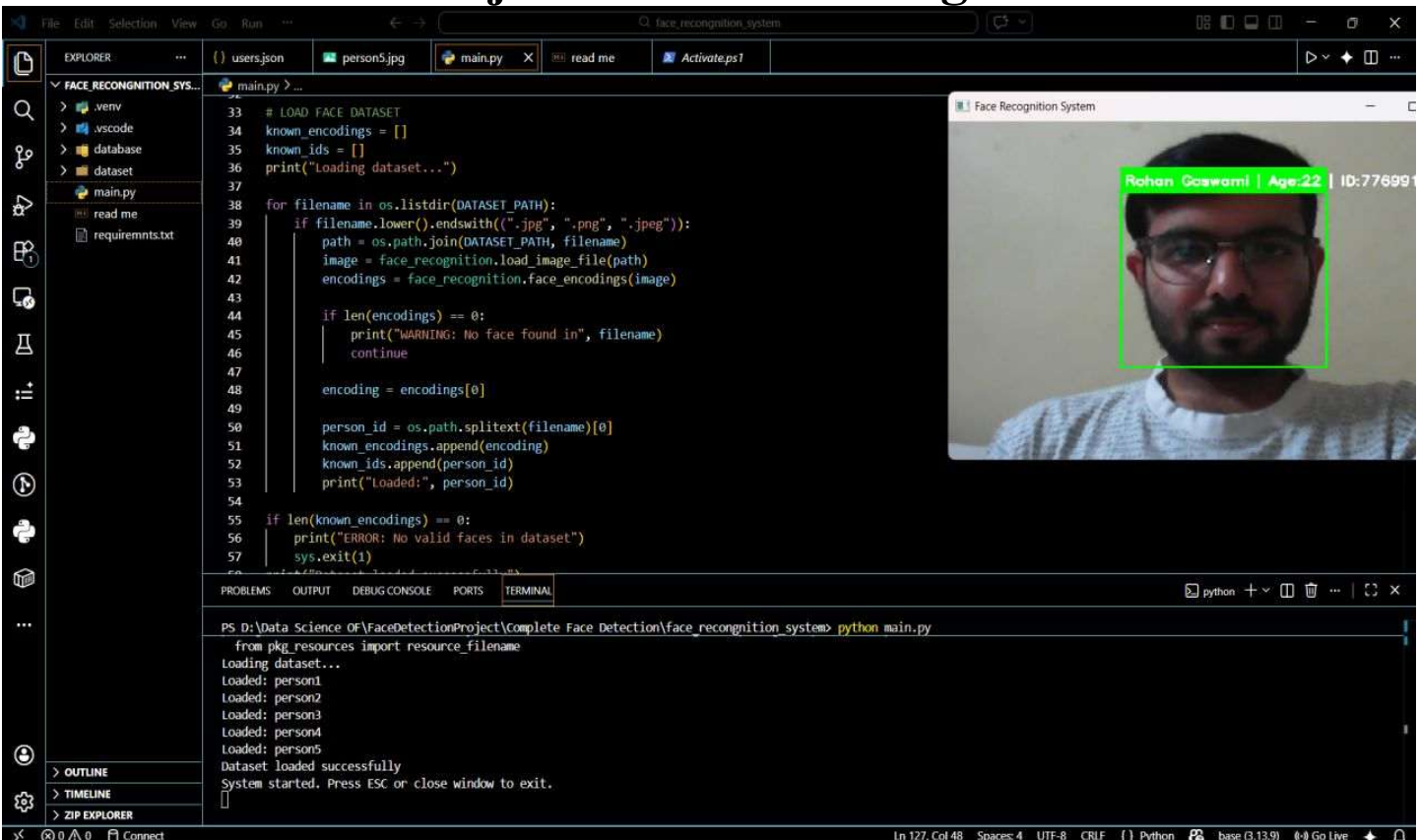
Project Structure



The screenshot shows the VS Code interface with the 'face_recognition_system' project open. The Explorer panel on the left displays the project structure, including a 'dataset' folder with five image files (person1.png to person5.jpg) and a 'main.py' file. The main editor shows the 'main.py' file with the following code:

```
60 # START CAMERA
61 video = cv2.VideoCapture(0)
62
63 if not video.isOpened():
64     print("ERROR: Cannot open camera")
65     sys.exit(1)
66
67 cv2.namedWindow("Face Recognition System", cv2.WINDOW_NORMAL)
68 print("System started. Press ESC or close window to exit.")
69
70 # MAIN LOOP
71 while True:
72     ret, frame = video.read()
73
74     if not ret:
75         print("Camera error")
76         break
77
78     # Resize for speed
79     small_frame = cv2.resize(frame, (0,0), fx=0.25, fy=0.25)
80     rgb_small = cv2.cvtColor(small_frame, cv2.COLOR_BGR2RGB)
81     face_locations = face_recognition.face_locations(rgb_small)
82     face_encodings = face_recognition.face_encodings(rgb_small, face_locations)
83
84     for encoding, location in zip(face_encodings, face_locations):
85         distances = face_recognition.face_distance(known_encodings, encoding)
86         best_index = np.argmin(distances)
87         best_distance = distances[best_index]
88
89         # Default values
90         name = "Unknown"
91         age = "-"
92         pid = "-"
93         color = (0,0,255) # red for unknown
94
95         # Only accept if distance is safe
96         if best_distance < MATCH_THRESHOLD:
97             person_key = known_ids[best_index]
```

Project Demo Working



The screenshot shows the VS Code interface with the 'face_recognition_system' project open. The Explorer panel on the left displays the project structure, including a 'dataset' folder with five image files (person1.png to person5.jpg) and a 'main.py' file. The main editor shows the 'main.py' file with the following code:

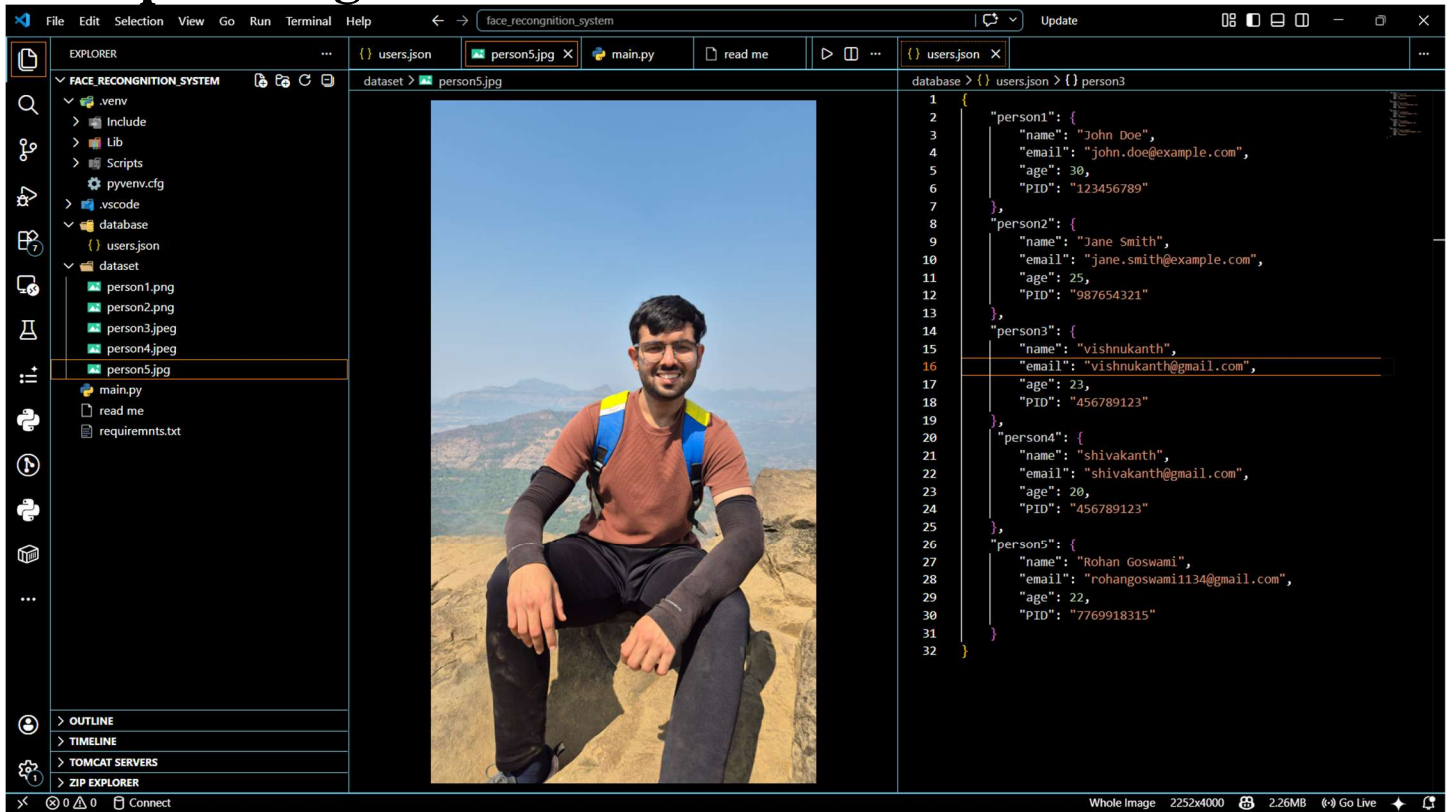
```
33 # LOAD FACE DATASET
34 known_encodings = []
35 known_ids = []
36 print("Loading dataset...")
37
38 for filename in os.listdir(DATASET_PATH):
39     if filename.lower().endswith((".jpg", ".png", ".jpeg")):
40         path = os.path.join(DATASET_PATH, filename)
41         image = face_recognition.load_image_file(path)
42         encodings = face_recognition.face_encodings(image)
43
44         if len(encodings) == 0:
45             print("WARNING: No face found in", filename)
46             continue
47
48         encoding = encodings[0]
49
50         person_id = os.path.splitext(filename)[0]
51         known_encodings.append(encoding)
52         known_ids.append(person_id)
53         print("Loaded:", person_id)
54
55 if len(known_encodings) == 0:
56     print("ERROR: No valid faces in dataset")
57     sys.exit(1)
```

The bottom panel shows the terminal output:

```
PS D:\Data Science OF\FaceDetectionProject\Complete Face Detection\face_recognition_system> python main.py
from pkg_resources import resource_filename
Loading dataset...
Loaded: person1
Loaded: person2
Loaded: person3
Loaded: person4
Loaded: person5
Dataset loaded successfully
System started. Press ESC or close window to exit.
```

On the right side of the terminal, there is a small window titled 'Face Recognition System' showing a video feed of a person's face. A green bounding box is drawn around the face, and text is overlaid on the image: 'Rohan Caswami | Age:22 | ID:776991'.

Input image dataset and Person Details in JSON



Visit Project at: <https://github.com/Rohan-1103/Face-Detection>

Internship Repository: <https://github.com/Rohan-1103/CodeIT-Internship>

CONCLUSION

The On-the-Job Training (OJT) provided a valuable opportunity to bridge the gap between academic learning and real-world application in the domain of Artificial Intelligence and Computer Vision. Through this internship, I gained practical exposure to building intelligent systems and applying machine learning concepts in a structured and professional environment.

1. Practical Implementation of Computer Vision Systems

The primary outcome of this internship was the successful development of a real-time face recognition system. This project enabled me to understand the complete pipeline of computer vision applications, including face detection, feature extraction, encoding, and identity matching. It strengthened my ability to translate theoretical AI concepts into a working system.

2. Understanding of Deep Learning-Based Feature Extraction

A significant learning outcome was gaining knowledge of how pre-trained deep learning models generate 128-dimensional facial embeddings. I developed an understanding of deep metric learning and how similarity-based approaches are used for recognition tasks instead of traditional classification models.

3. Data Handling and System Integration

The internship enhanced my ability to work with structured datasets and integrate them with real-time systems. By using JSON-based metadata for identity mapping, I learned how to manage data efficiently and ensure consistency between stored information and real-time predictions.

4. Real-Time System Optimization

Another key outcome was learning how to optimize applications for real-time performance. Techniques such as frame downscaling and efficient computation helped in achieving smooth execution on CPU-based systems, improving both performance and responsiveness.

5. Problem-Solving and System Reliability

The internship improved my problem-solving skills by exposing me to real-world challenges such as handling missing data, multiple detections, and system failures. Implementing validation checks and error handling mechanisms enhanced the robustness and reliability of the system.

6. Professional Development and Industry Readiness

Beyond technical skills, this OJT experience helped me develop a disciplined approach towards project development, documentation, and debugging. It improved my ability to work independently, manage tasks effectively, and think critically while solving problems, preparing me for future roles in the AI and software industry.

In conclusion, this internship was a significant step in my learning journey, enabling me to gain hands-on experience in Artificial Intelligence and Computer Vision. It has strengthened my technical foundation and provided me with the confidence and skills required to contribute effectively to real-world AI-driven applications.